Substance Over Style: Mitigating Tone Bias in Retrieval-Augmented Generation across Conflicting Evidence

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Problem and Motivation: Retrieval-augmented generation (RAG) systems improve the factual grounding of language model outputs by conditioning generation on retrieved documents [1]. In practice, however, retrieved documents frequently conflict: two sources may make opposing factual claims on the same query. Standard RAG pipelines have no mechanism for explicitly resolving such conflicts: the generator receives all retrieved content and produces an answer by implicitly weighting competing claims, with no transparency into which evidence drove the output. Debate-augmented RAG addresses this through structured argumentation: each retrieved document is assigned a dedicated agent that argues for its answer, and a judge agent arbitrates by selecting the better-supported position [1]. This improves robustness over standard RAG under ambiguous retrieval, but implicitly assumes the judge selects winners on the basis of substance: the factual accuracy, evidential grounding, and source credibility of the underlying documents. We identify a threat to this assumption that has not been studied in a RAG setting: a well-functioning system should favor substance over style, but a document asserting "it is well-established that X" may dominate one stating "evidence suggests X may be the case," even when the latter is more credible and factually correct. This form-over-substance failure is particularly dangerous in medicine, law, and finance, where hedging language is a mark of rigor, not weakness. We study whether this failure occurs in standard and debate-augmented RAG, and introduce Evidence-Anchored Judge Prompting (EAJP), a claim-level intervention designed to restore substance-first judgment under conflicting evidence.

Related Work: Wang et al. introduce RAMDocs, a benchmark modeling ambiguity, noise, and misinformation in retrieved documents, and propose MADAM-RAG, a multi-agent debate system where each document is represented by a dedicated agent that argues for its answer across multiple rounds before an aggregator resolves the conflict [1]. Debate improves robustness over single-pass RAG under noisy and misleading retrieval, but the aggregator is treated as a reliable arbitrator and the influence of document tone on judge decisions is not analyzed. Tang et al. propose RA-RAG, which estimates source reliability via cross-document consistency before aggregation, addressing credibility at the retrieval level but not how rhetorical style interacts with credibility signals at the generation or judgment stage [2]. A growing body of work shows that LLM judges treat stylistic features such as verbosity, authority, and confident tone as proxies for quality independent of correctness [3, 4, 5]. Agarwal and Khanna demonstrate this effect directly in multi-agent debate, showing that a confident argument for a false claim reliably defeats a correct but neutrally-framed one [6]. Saeidi et al. demonstrate that inserting rhetorical strategies, such appeals to authority and flattery, causes LLM judges to assign significantly higher scores resulting in style trumping correctness in structured evaluation tasks [5]. Complementing this, recent work finds that judges not only pick the wrong answer when misled by rhetoric-they do so with miscalibrated high certainty, compounding the reliability problem [7]. Together, these findings motivate the question of whether tone bias of this kind affects RAG systems under conflicting evidence, and whether a claim-level prompting intervention can restore substance-first judgment, a question that remains unexplored. EAJP prompts a judge to identify the most relevant claim in each document, locate the supporting evidence, assess cross-document consistency and source credibility, and produce a final answer grounded in evidence rather than rhetorical confidence.

Research Questions: We study two questions: (1) Does standard RAG and MADAM-RAG exhibit tone bias under conflicting evidence, systematically favoring assertively-written documents over better-evidenced but neutrally-phrased ones? and (2) Does EAJP reduce this bias?

Approach and Experimental Plan: We compare three systems: standard RAG, MADAM-RAG [1], and EAJP, using GPT-5.2 and Claude Sonnet 4.6 as judge models. We evaluate these systems on two existing benchmarks: AmbigDocs, which pairs queries with documents supporting genuinely different interpretations of the same question, and a filtered subset of FaithEval, which contains instances where some retrieved documents contradict a known correct answer [1] and a new Tone-Controlled Benchmark (TCB) which we construct from them. We retain only cases where at least one document supports the correct answer, ensuring the task requires selecting the right claim rather than detecting its absence. We also construct a Tone-Controlled Benchmark (TCB) by rewriting retrieved documents in assertive and neutral registers using Gemini 2.5 Pro, which we verify preserves the meaning of each claim. TCB targets conditions where the factually correct document from a high-credibility source is written neutrally while the incorrect document from a low-credibility source is written assertively. Our primary metrics are substance alignment rate, which we define as the proportion of trials where the final answer reflects the correct and higher-credibility claim, and tone bias rate, which we define as the proportion where the system favors the assertive but incorrect claim over the better-evidenced neutral one.

References

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AI Usage Disclosure

AI tools such as Perplexity were used for rewriting select sections of this proposal and grammar checking. The core intellectual contributions of this proposal, the project idea, selection of prior work, proposed approach, and research plan, were developed without the use of AI tools.